DATA ANALYTICS LAB SHEET 4

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Experiment No. 4

Advanced Data Visualization using Matplotlib and Seaborn

AIM

To create and analyze various data visualizations using Python libraries such as Matplotlib and Seaborn in order to identify trends, patterns, correlations, and insights from real-world datasets.

DATASET

Superstore Sales Dataset.

PURPOSE

Data visualization transforms raw data into graphical representations, making it easier to understand complex datasets and communicate insights effectively.

1. Import Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")

print("Libraries imported successfully.")
```

```
Libraries imported successfully.
```

2. Load the Dataset

```
# Load the Superstore dataset from a verified public GitHub repository
url =
"https://raw.githubusercontent.com/leonism/sample-superstore/master/
data/superstore.csv"
```

```
df = pd.read_csv(url)

print("Dataset loaded successfully.")
print("Dataset shape:", df.shape)
display(df.head())
```

```
Dataset loaded successfully.
Dataset shape: (10800, 21)
```

```
{"type":"dataframe"}
```

3. Inspect and Prepare the Dataset

```
print("Column names:")
print(df.columns.tolist())

print("\nDataset information:")
df.info()

print("\nSummary statistics:")
display(df.describe(include="all").T)

# Convert date columns where available
for col in ["Order Date", "Ship Date"]:
    if col in df.columns:
        df[col] = pd.to_datetime(df[col], errors="coerce")

print("\nMissing values:")
display(df.isnull().sum().sort_values(ascending=False).head(15))
```

```
Column names:
['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode',
'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State',
'Postal Code', 'Region', 'Product ID', 'Category', 'Sub-Category',
'Product Name', 'Sales', 'Quantity', 'Discount', 'Profit']

Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10800 entries, 0 to 10799
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         10800 non-null  object
 1   Order ID       10800 non-null  object
 2   Order Date     9994 non-null   object
 3   Ship Date      9994 non-null   object
 4   Ship Mode      9994 non-null   object
 5   Customer ID    9994 non-null   object
 6   Customer Name  9994 non-null   object
 7   Segment        9994 non-null   object
```

```
 8   Country        9994 non-null   object
 9   City           9994 non-null   object
 10  State          9994 non-null   object
 11  Postal Code    9983 non-null   float64
 12  Region         9994 non-null   object
 13  Product ID     9994 non-null   object
 14  Category       9994 non-null   object
 15  Sub-Category   9994 non-null   object
 16  Product Name   9994 non-null   object
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   float64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: float64(5), object(16)
memory usage: 1.7+ MB

Summary statistics:

{"summary":"{\n  \"name\": \"display(df\",\n  \"rows\": 21,\n
\"fields\": [\n    {\n      \"column\": \"count\",\n
\"properties\": {\n        \"dtype\": \"date\",\n        \"min\":
9983.0,\n        \"max\": \"10800\",\n        \"num_unique_values\":
3,\n        \"samples\": [\n          \"10800\",\n          \"9994\",\
n          9983.0\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"unique\",\n      \"properties\": {\n        \"dtype\": \"date\",\n
\"min\": 1,\n        \"max\": 10001,\n        \"num_unique_values\":
13,\n        \"samples\": [\n          17,\n          49,\n
10001\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"top\",\n      \"properties\": {\n        \"dtype\": \"string\",\n
\"num_unique_values\": 16,\n        \"samples\": [\n
\"Yes\",\n          \"CA-2018-100111\",\n          \"WB-21850\"\n
],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n
}\n    },\n    {\n      \"column\": \"freq\",\n      \"properties\":
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\"max\": \"9994\",\n        \"num_unique_values\": 15,\n
\"samples\": [\n          \"2001\",\n          \"19\",\n
\"800\"\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"mean\",\n      \"properties\": {\n        \"dtype\": \"date\",\n
\"min\": 0.15620272163297977,\n        \"max\": 55245.23329660423,\n
\"num_unique_values\": 5,\n        \"samples\": [\n
229.85800083049827,\n          28.65689630778467,\n
3.789573744246548\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"std\",\n      \"properties\": {\n        \"dtype\": \"date\",\n
\"min\": 0.2064519678256984,\n        \"max\": 32038.71595549829,\n
\"num_unique_values\": 5,\n        \"samples\": [\n
623.2451005086818,\n          234.26010769095757,\n
```

2.2251096911414274\n   ],\n   \"semantic_type\": \"\",\n \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"min\",\n   \"properties\": {\n   \"dtype\": \"date\",\n \"min\": -6599.978,\n   \"max\": 1040.0,\n \"num_unique_values\": 5,\n   \"samples\": [\n   0.444,\n -6599.978,\n   1.0\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n \"column\": \"25%\",\n   \"properties\": {\n   \"dtype\": \"date\",\n   \"min\": 0.0,\n   \"max\": 23223.0,\n \"num_unique_values\": 5,\n   \"samples\": [\n   17.28,\n 1.72875,\n   2.0\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n \"column\": \"50%\",\n   \"properties\": {\n   \"dtype\": \"date\",\n   \"min\": 0.2,\n   \"max\": 57103.0,\n \"num_unique_values\": 5,\n   \"samples\": [\n 54.489999999999995,\n   8.6665,\n   3.0\n   ],\n \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"75%\",\n   \"properties\": {\n \"dtype\": \"date\",\n   \"min\": 0.2,\n   \"max\": 90008.0,\n   \"num_unique_values\": 5,\n   \"samples\": [\n 209.94,\n   29.364,\n   5.0\n   ],\n \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"max\",\n   \"properties\": {\n \"dtype\": \"date\",\n   \"min\": 0.8,\n   \"max\": 99301.0,\n   \"num_unique_values\": 5,\n   \"samples\": [\n 22638.48,\n   8399.976,\n   14.0\n   ],\n \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   }\n]\n}","type":"dataframe"}

Missing values:

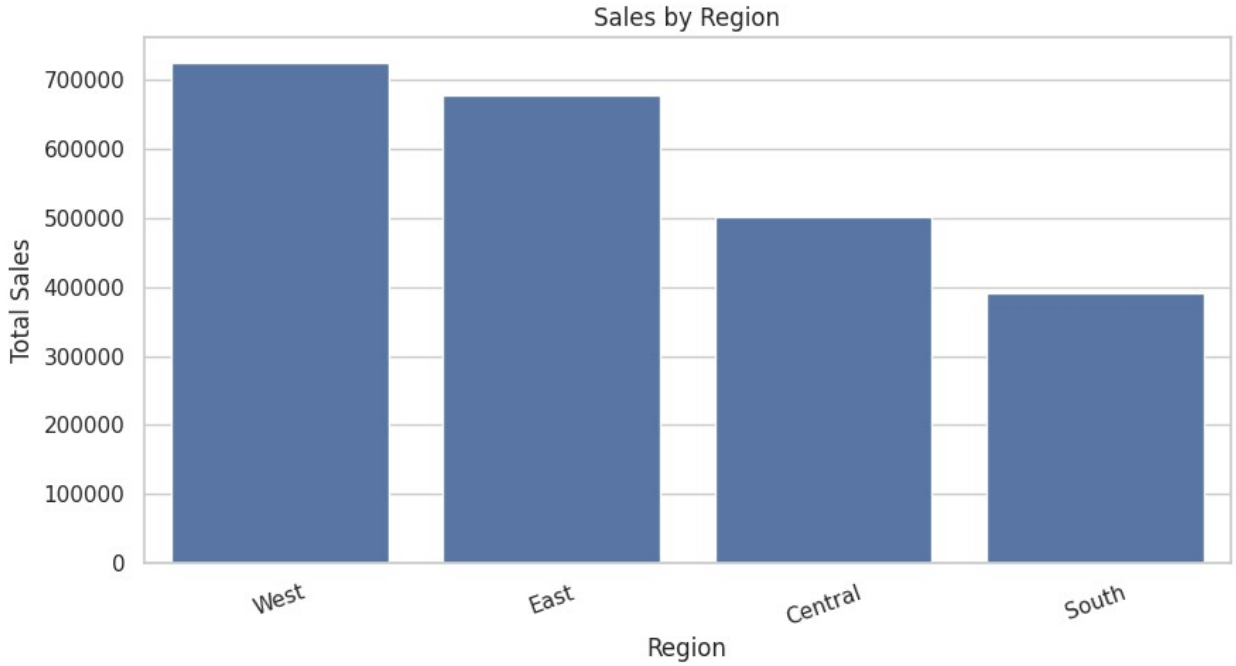
Postal Code	817
Ship Mode	806
Ship Date	806
Customer ID	806
Order Date	806
Product ID	806
Category	806
Segment	806
Customer Name	806
Country	806
City	806
State	806
Region	806
Sales	806
Quantity	806

dtype: int64

4. Bar Chart - Sales by Region

```
region_sales = df.groupby("Region")
["Sales"].sum().sort_values(ascending=False)

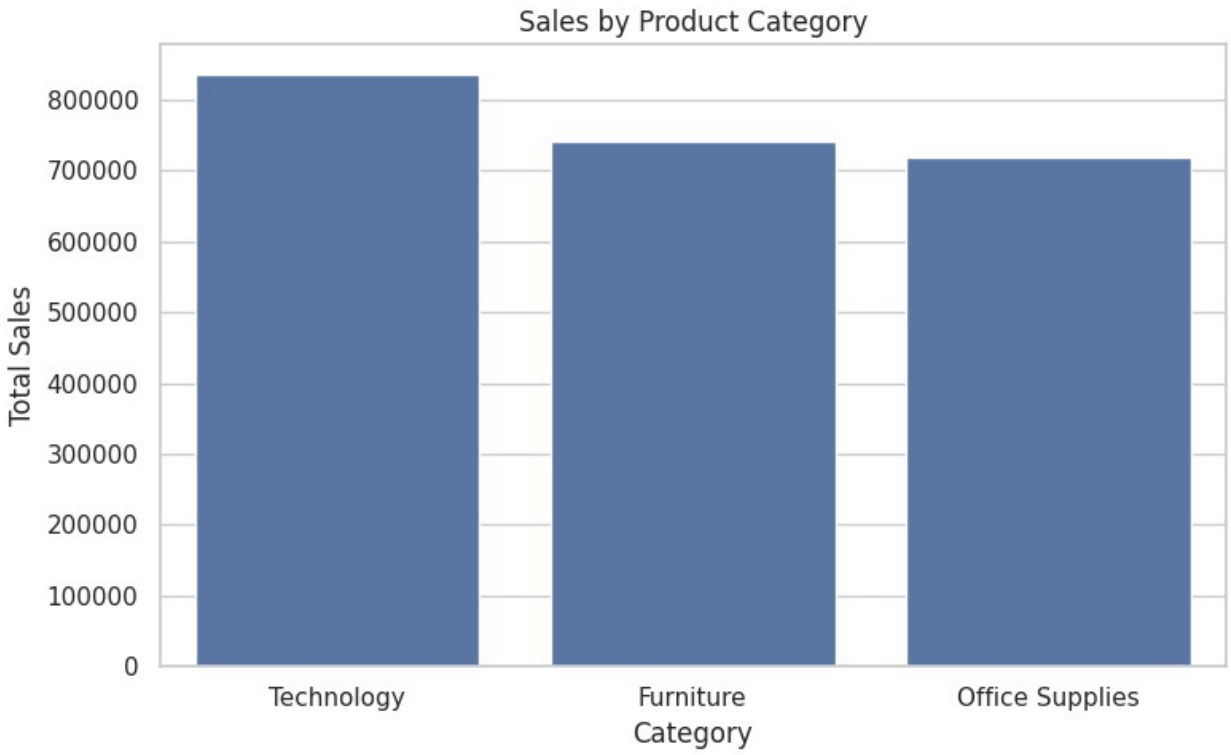
plt.figure(figsize=(9,5))
sns.barplot(x=region_sales.index, y=region_sales.values)
plt.title("Sales by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```



5. Bar Chart - Sales by Product Category

```
category_sales = df.groupby("Category")
["Sales"].sum().sort_values(ascending=False)

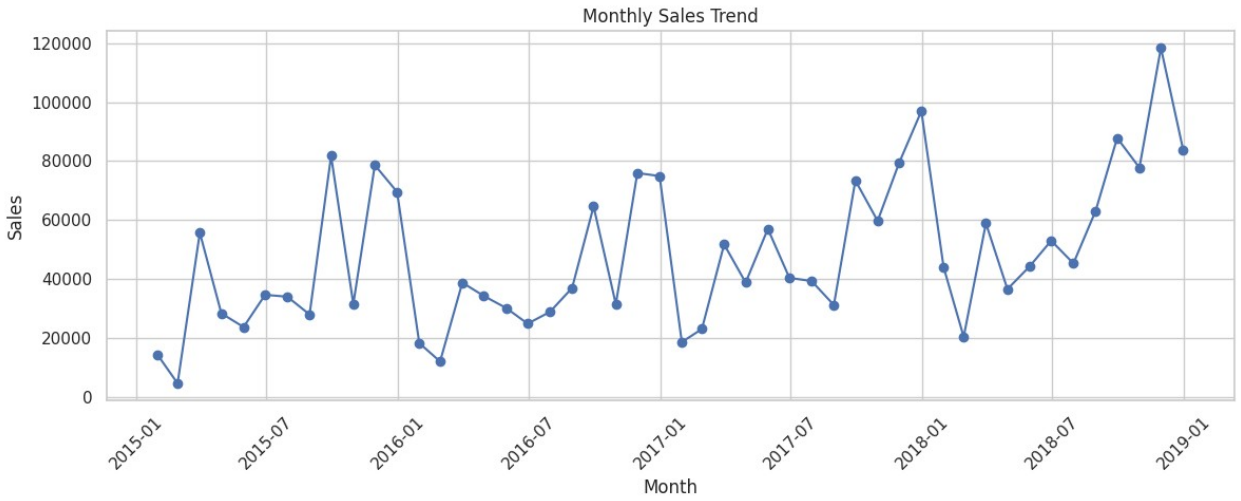
plt.figure(figsize=(8,5))
sns.barplot(x=category_sales.index, y=category_sales.values)
plt.title("Sales by Product Category")
plt.xlabel("Category")
plt.ylabel("Total Sales")
plt.tight_layout()
plt.show()
```



6. Line Chart - Sales Trend Over Time

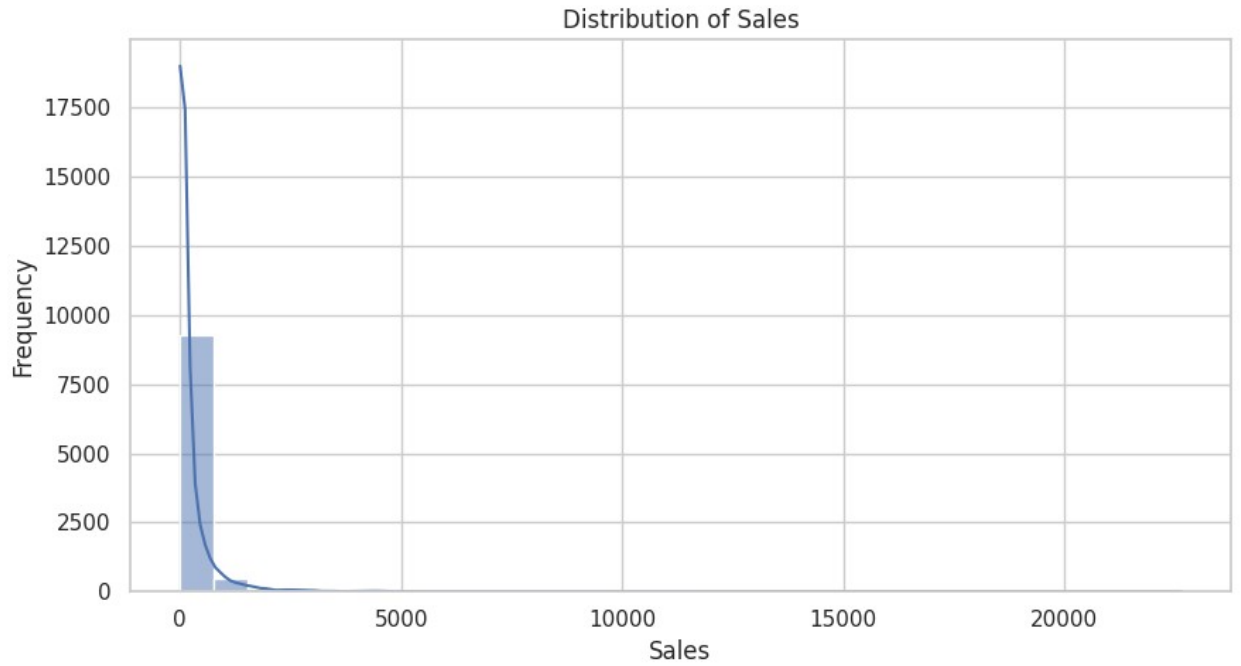
```
if "Order Date" in df.columns:
    monthly_sales = (
        df.dropna(subset=["Order Date"])
          .set_index("Order Date")["Sales"]
          .resample("ME")
          .sum()
    )

    plt.figure(figsize=(12,5))
    plt.plot(monthly_sales.index, monthly_sales.values, marker="o")
    plt.title("Monthly Sales Trend")
    plt.xlabel("Month")
    plt.ylabel("Sales")
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()
```



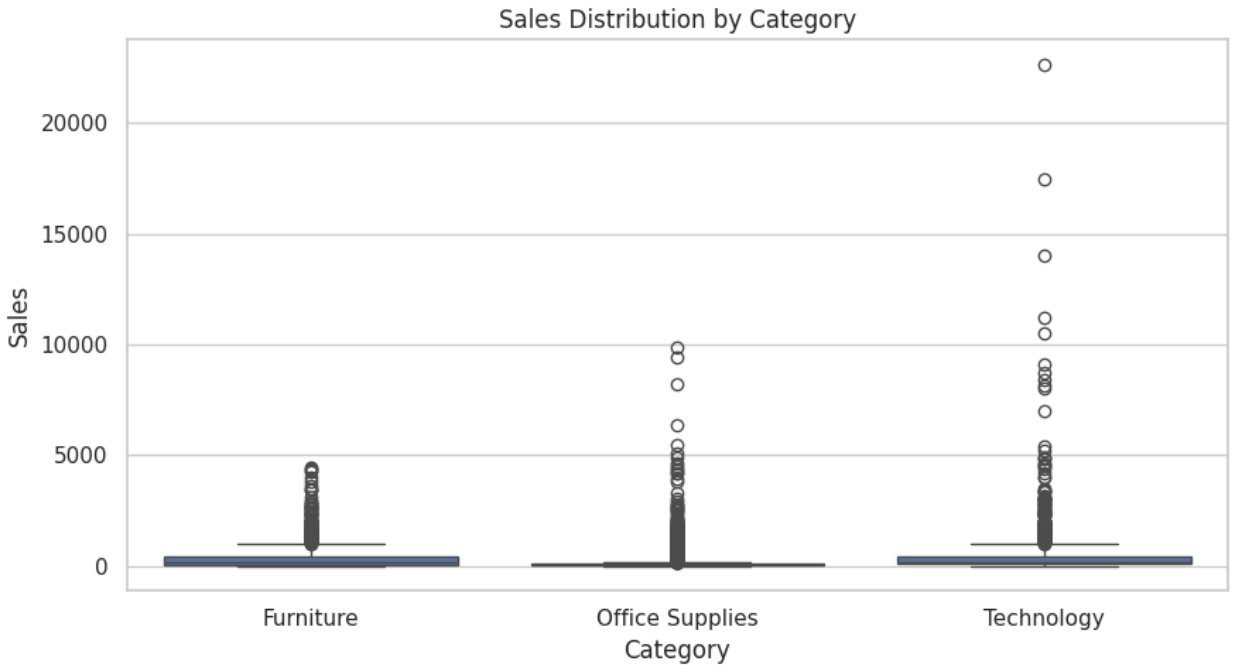
7. Histogram - Distribution of Sales

```
plt.figure(figsize=(9,5))
sns.histplot(df["Sales"].dropna(), bins=30, kde=True)
plt.title("Distribution of Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```



8. Box Plot - Sales by Category

```
plt.figure(figsize=(9,5))
sns.boxplot(data=df, x="Category", y="Sales")
plt.title("Sales Distribution by Category")
plt.xlabel("Category")
plt.ylabel("Sales")
plt.tight_layout()
plt.show()
```



9. Scatter Plot - Sales vs Profit

```
plt.figure(figsize=(9,6))
sns.scatterplot(
    data=df,
    x="Sales",
    y="Profit",
    hue="Category",
    alpha=0.7
)
plt.title("Sales vs Profit")
plt.xlabel("Sales")
plt.ylabel("Profit")
plt.tight_layout()
plt.show()
```

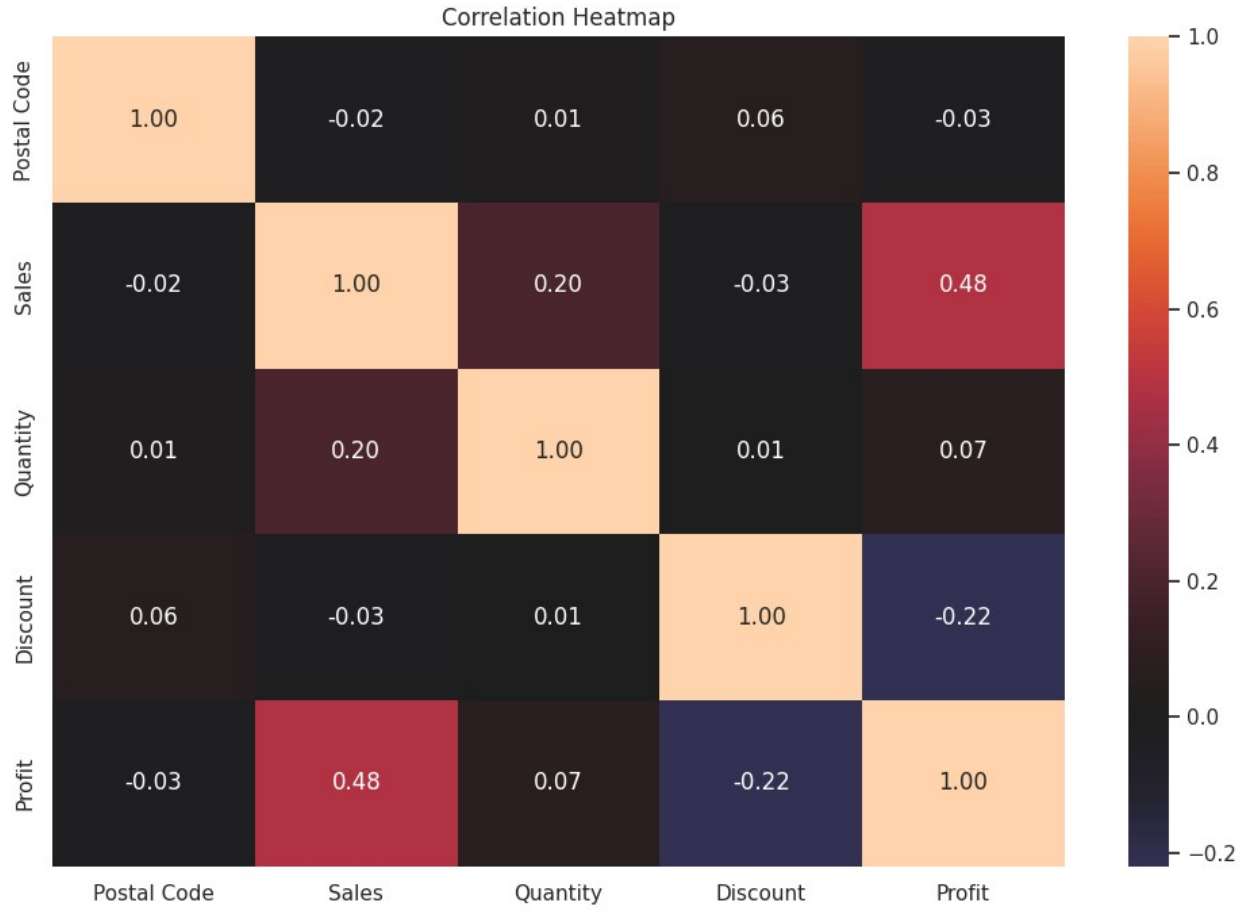


10. Correlation Heatmap

```
numeric_df = df.select_dtypes(include=np.number)

correlation = numeric_df.corr()

plt.figure(figsize=(10,7))
sns.heatmap(
    correlation,
    annot=True,
    fmt=".2f",
    center=0
)
plt.title("Correlation Heatmap")
plt.tight_layout()
plt.show()
```



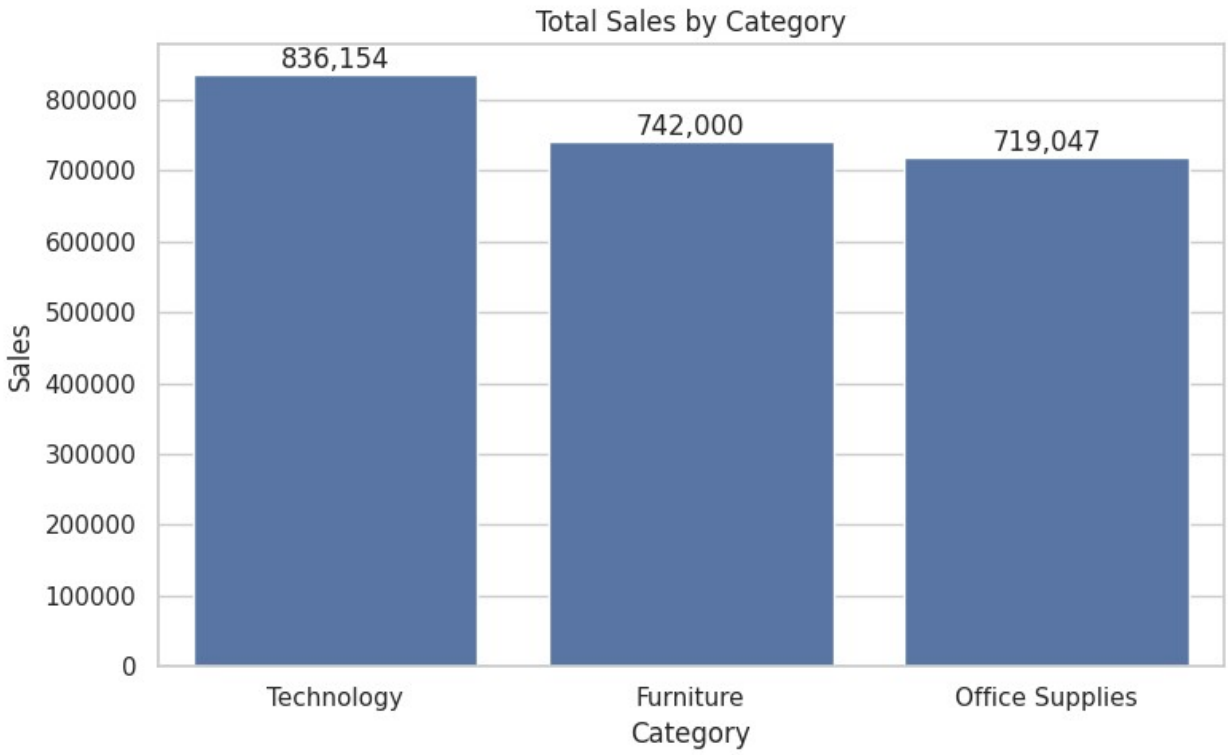
11. Customized Visualization with Annotations

```
top_categories = (
    df.groupby("Category")["Sales"]
      .sum()
      .sort_values(ascending=False)
)

plt.figure(figsize=(8,5))
ax = sns.barplot(x=top_categories.index, y=top_categories.values)

for i, value in enumerate(top_categories.values):
    ax.text(i, value, f"{value:,.0f}", ha="center", va="bottom")

plt.title("Total Sales by Category")
plt.xlabel("Category")
plt.ylabel("Sales")
plt.tight_layout()
plt.show()
```



12. Sales and Profit by Region

```
region_summary = (
    df.groupby("Region")[["Sales", "Profit"]]
      .sum()
      .sort_values("Sales", ascending=False)
)

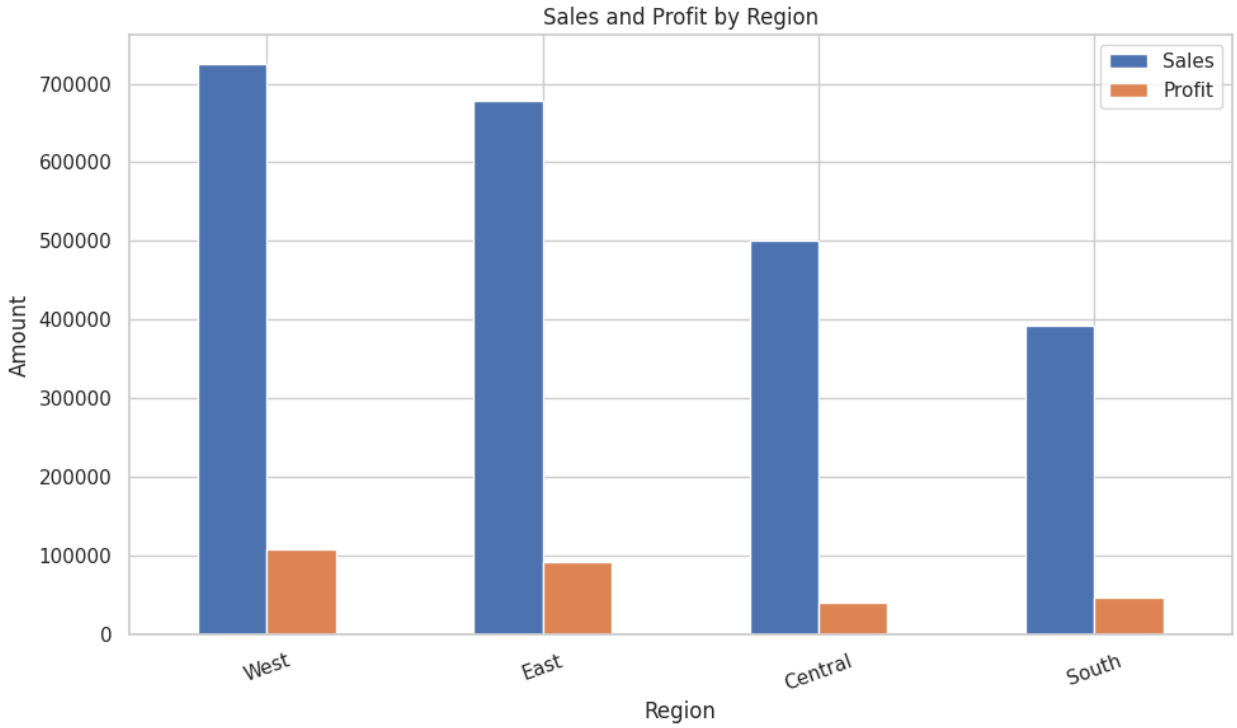
display(region_summary)

region_summary.plot(
    kind="bar",
    figsize=(10,6)
)

plt.title("Sales and Profit by Region")
plt.xlabel("Region")
plt.ylabel("Amount")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```


```
{"summary":"{\n  \"name\": \"region_summary\",\n  \"rows\": 4,\n
\"fields\": [\n    {\n      \"column\": \"Region\",\n
\"properties\": {\n        \"dtype\": \"string\",\n
\"num_unique_values\": 4,\n        \"samples\": [\n
\"East\",\n          \"South\",\n          \"West\"\n        ],\n
```

\"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"Sales\",\n   \"properties\": {\n   \"dtype\": \"number\",\n   \"std\": 155389.35801127684,\n   \"min\": 391721.905,\n   \"max\": 725457.8245,\n   \"num_unique_values\": 4,\n   \"samples\": [\n   678781.24,\n   391721.905,\n   725457.8245\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"Profit\",\n   \"properties\": {\n   \"dtype\": \"number\",\n   \"std\": 33601.93579497421,\n   \"min\": 39706.3625,\n   \"max\": 108418.4489,\n   \"num_unique_values\": 4,\n   \"samples\": [\n   91522.78,\n   46749.4303,\n   108418.4489\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   }\n]\n}","type":"dataframe","variable_name":"region_summary"}

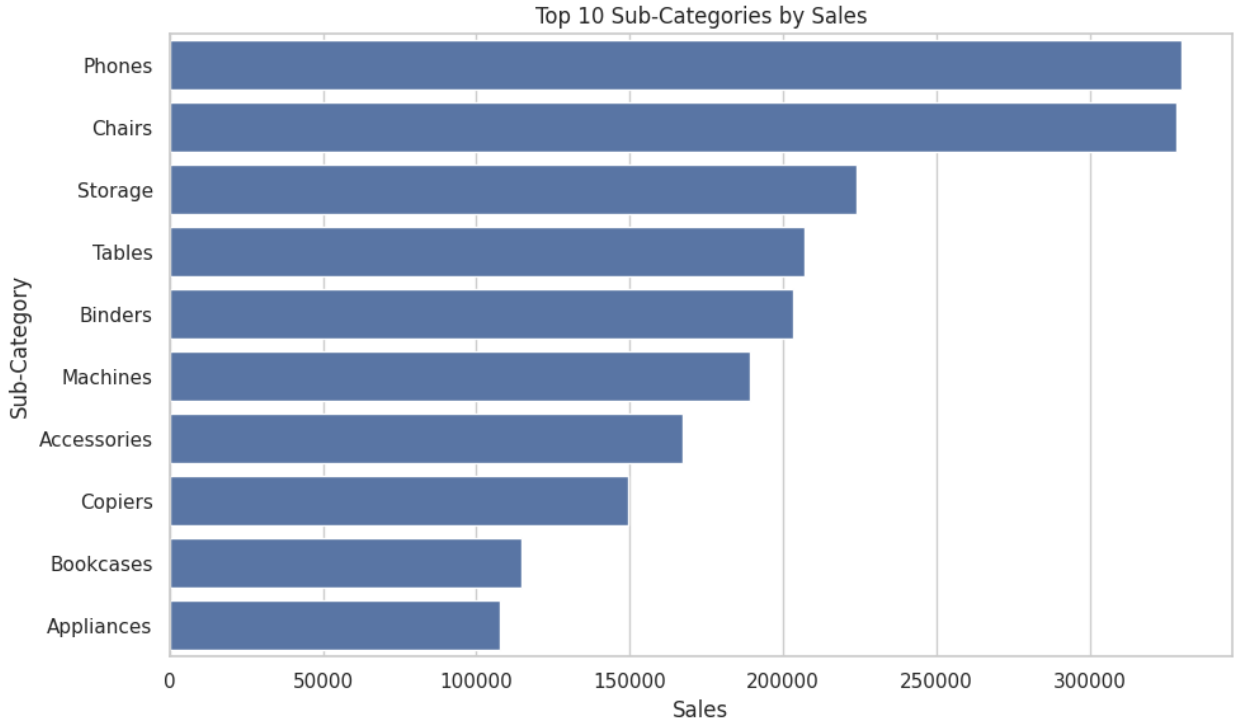


13. Top 10 Sub-Categories by Sales

```
top_subcategories = (
    df.groupby("Sub-Category")["Sales"]
      .sum()
      .sort_values(ascending=False)
      .head(10)
)

plt.figure(figsize=(10,6))
sns.barplot(
    x=top_subcategories.values,
```

```
    y=top_subcategories.index
)
plt.title("Top 10 Sub-Categories by Sales")
plt.xlabel("Sales")
plt.ylabel("Sub-Category")
plt.tight_layout()
plt.show()
```



14. Business Insights

```
print("KEY BUSINESS INSIGHTS")
print("----------------------")

best_region = region_sales.idxmax()
best_category = category_sales.idxmax()

print(f"1. The region with the highest total sales is: {best_region}")
print(f"2. The product category with the highest total sales is:
{best_category}")

if "Profit" in df.columns:
    most_profitable_region = df.groupby("Region")
["Profit"].sum().idxmax()
    print(f"3. The most profitable region based on total profit is:
{most_profitable_region}")

print("4. The histogram shows the distribution and spread of sales
```

```
values.")
print("5. The box plot helps identify variation and potential outliers
in sales.")
print("6. The scatter plot shows the relationship between sales and
profit.")
print("7. The correlation heatmap shows relationships among numerical
variables.")
```

```
KEY BUSINESS INSIGHTS
----------------------
1. The region with the highest total sales is: West
2. The product category with the highest total sales is: Technology
3. The most profitable region based on total profit is: West
4. The histogram shows the distribution and spread of sales values.
5. The box plot helps identify variation and potential outliers in
sales.
6. The scatter plot shows the relationship between sales and profit.
7. The correlation heatmap shows relationships among numerical
variables.
```

RESULT

Advanced data visualization was successfully performed using Matplotlib and Seaborn. Bar charts, line charts, histograms, box plots, scatter plots, and a correlation heatmap were created and interpreted. The visualizations were customized with titles, labels, annotations, legends, and suitable color palettes to identify trends, patterns, relationships, and business insights.

Answers to Questions

Q1. Why is data visualization considered an essential component of data analytics?

Data visualization converts complex numerical information into graphical form. It makes patterns, trends, outliers, and relationships easier to identify and helps communicate analytical findings clearly.

Q2. Differentiate between Matplotlib and Seaborn. Which library is more suitable for statistical visualizations and why?

Matplotlib is a general-purpose Python visualization library that provides detailed control over plots. Seaborn is built on Matplotlib and provides a higher-level interface with attractive statistical plots and simpler syntax. Seaborn is generally more convenient for statistical visualizations because it provides functions for distributions, categorical analysis, regression, and relationships.

Q3. Which type of chart would you choose to compare sales across different regions? Justify your answer.

A bar chart is suitable for comparing sales across regions because each region is represented by a separate bar, making differences in sales easy to compare.

Q4. What information can be obtained from a histogram and a box plot?

A histogram shows the distribution, frequency, spread, and shape of a numerical variable. A box plot shows the median, quartiles, overall spread, and potential outliers.

Q5. Explain the purpose of a scatter plot. How does it help in identifying relationships between variables?

A scatter plot displays pairs of numerical observations. It helps identify the direction and strength of a relationship, clusters, trends, and unusual observations between two variables.

Q6. What is a correlation heatmap? How can it assist in feature selection for machine learning?

A correlation heatmap is a graphical representation of correlation coefficients among numerical variables. It can help identify highly correlated features and possible redundancy. Features with very high correlation may provide similar information and can be considered during feature selection.

Q7. Why is chart customization important in data visualization?

Titles, axis labels, legends, annotations, and suitable colors make charts easier to understand. Good customization improves readability and ensures that the main analytical message is communicated clearly.

Q8. What factors should be considered while selecting an appropriate visualization for a dataset?

The type of variables, analytical objective, number of categories, time dimension, relationship being studied, audience, and amount of data should be considered. The selected chart should clearly represent the intended information without unnecessary complexity.

Q9. How can misleading visualizations affect business decision-making? Give a real-world example.

Misleading scales, inappropriate chart types, selective data, or missing context can make differences appear larger or smaller than they actually are. For example, starting a bar chart's vertical axis above zero can exaggerate a small difference in sales between two regions and may lead managers to make an incorrect decision.

Q10. Explain how data visualization supports storytelling and business intelligence in organizations.

Data visualization turns analytical results into a clear visual story. Dashboards and charts help managers monitor KPIs, identify trends, compare performance, detect problems, and communicate insights. This supports faster and more informed business decisions.